

Surrogate-model comparison — heteroscedastic mixed-variable test problems

Separate GPs / Categorical GP / Standard LVGP (MATLAB) / H-LVGP (MATLAB)

Legend. Metrics on a ground-truth test set (200-pt grid per level for 1-D; 1024 Sobol points per level for Rastrigin 6-D): MAE and RRMSE (surface accuracy), 90% Gneiting–Raftery interval score IS (accuracy + calibration), and 90% coverage (target 0.90; higher is *not* better). **Bold on green** = best model in the comparison group (winners are recomputed within each version’s model set). Budgets n are TOTAL design points; designs are SLHD with n_{rep} replicates per point. Single seed.

Version 1 — all four models

Table 1: Mean-over-levels metrics for every problem $\times$ budget $\times$ model (seed 1). Bold on green = best model in each (problem, budget, n_{rep}) block per metric (MAE/RRMSE/IS: lower; coverage: closest to 0.90).

Problem	n	Model	$n_{\text{rep}} = 10$				$n_{\text{rep}} = 3$			
			MAE $\downarrow$	RRMSE $\downarrow$	IS $\downarrow$	Cov. $\rightarrow .9$	MAE $\downarrow$	RRMSE $\downarrow$	IS $\downarrow$	Cov. $\rightarrow .9$
Branin	10	Separate GPs	26.17	0.8475	360.3	0.6680	26.33	0.8600	349.7	0.6860
		Categorical GP	24.54	0.7582	173.5	0.9420	24.66	0.7623	174.4	0.9420
		Standard LVGP	24.71	0.8273	332.6	0.3300	35.03	1.280	469.1	0.2670
		H-LVGP	43.37	1.663	589.7	0.3540	34.96	1.279	447.5	0.3460
	25	Separate GPs	9.591	0.3852	69.31	0.6860	9.594	0.3704	66.56	0.6710
		Categorical GP	3.582	0.1369	23.57	0.8300	3.611	0.1288	21.75	0.7900
		Standard LVGP	2.553	0.1385	19.18	0.3840	2.769	0.1641	21.31	0.4950
		H-LVGP	2.094	0.0899	11.29	0.9920	1.908	0.1196	13.55	0.9900
	40	Separate GPs	2.980	0.1586	23.32	0.8850	3.201	0.1712	24.54	0.8680
		Categorical GP	0.8582	0.0338	6.284	1.000	1.214	0.0535	5.889	0.9270
		Standard LVGP	1.880	0.0736	19.88	0.1150	1.382	0.0627	19.12	0.2990
		H-LVGP	0.6627	0.0271	9.348	1.000	1.384	0.0555	8.583	0.9800
Six-hump camel	8	Separate GPs	0.9429	1.122	9.221	0.6462	0.9168	1.084	8.743	0.6238
		Categorical GP	0.5557	0.6229	3.890	0.7225	0.5638	0.6158	3.376	0.7613
		Standard LVGP	0.6343	0.7084	8.216	0.2200	0.6275	0.6984	7.442	0.2562
		H-LVGP	0.6319	0.7042	7.014	0.4187	0.6297	0.6964	6.341	0.4050
	20	Separate GPs	0.2411	0.3526	1.524	0.9550	0.2487	0.3837	1.640	0.9575
		Categorical GP	0.1640	0.2511	1.118	0.8525	0.1542	0.2473	0.9751	0.8738
		Standard LVGP	0.2097	0.3159	2.332	0.2725	0.2889	0.4643	4.071	0.1500
		H-LVGP	0.1821	0.2838	1.140	0.9187	0.2537	0.4077	2.191	0.7638
	32	Separate GPs	0.1883	0.3305	1.636	0.8600	0.1984	0.3472	1.804	0.8137
		Categorical GP	0.0845	0.1288	0.5680	0.9287	0.0870	0.1212	0.5249	0.8700
		Standard LVGP	0.1400	0.2662	1.675	0.3425	0.1399	0.2376	1.747	0.2088
		H-LVGP	0.1073	0.2081	0.6652	0.9938	0.0907	0.1342	0.9590	0.9800
Rastrigin 6-D	80	Separate GPs	19.58	0.8149	141.1	0.6479	19.73	0.8203	143.5	0.6438
		Categorical GP	19.47	0.8178	133.7	0.6729	19.78	0.8300	141.5	0.6475
		Standard LVGP	18.68	0.7784	117.5	0.7393	18.85	0.7858	120.8	0.7258
		H-LVGP	18.48	0.7700	114.1	0.7490	18.66	0.7777	117.7	0.7373
	160	Separate GPs	19.02	0.7907	111.9	0.7898	19.64	0.8195	118.8	0.7659
		Categorical GP	19.33	0.7995	112.8	0.7725	19.44	0.8043	114.0	0.7690
		Standard LVGP	19.54	0.8134	113.7	0.7996	19.78	0.8245	115.6	0.7942
		H-LVGP	19.56	0.8154	113.4	0.8030	19.65	0.8189	113.7	0.7991

Table 2: Branin — per-level metrics, $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
10	$x_2 = 15$	25.65	0.4851	323.4	1.000	28.58	0.5221	103.9	1.000	16.74	0.3861	112.3	0.6950	52.82	0.9546	864.8	0.2250
	$x_2 = 2$	28.17	0.9843	305.5	0.8550	25.07	0.8346	210.5	0.8800	25.42	0.7236	238.0	0.4700	29.90	0.9110	200.5	0.6050
	$x_2 = 8$	15.29	1.050	173.9	0.4350	13.48	0.8945	109.9	1.000	18.17	1.243	267.8	0.1350	53.29	3.661	855.1	0.1900
	$x_2 = 0$	47.34	1.116	897.1	0.0800	42.37	0.9837	329.6	0.8450	47.84	1.118	830.0	0.0000	40.37	0.9643	427.8	0.5600
	$x_2 = 10$	14.40	0.6024	101.4	0.9700	13.19	0.5559	113.9	0.9850	15.40	0.6668	214.7	0.3500	40.46	1.824	600.5	0.1900
	<i>Mean</i>	26.17	0.8475	360.3	0.6680	24.54	0.7582	173.5	0.9420	24.71	0.8273	332.6	0.3300	43.37	1.663	589.7	0.3540
25	$x_2 = 15$	5.808	0.1166	37.32	0.5700	3.739	0.0807	19.29	0.8250	3.470	0.0899	38.02	0.5200	3.163	0.0926	16.13	1.000
	$x_2 = 2$	14.27	0.4126	110.1	0.7850	3.752	0.1410	26.61	0.8350	1.699	0.0535	11.19	0.3800	1.618	0.0382	7.714	1.000
	$x_2 = 8$	6.595	0.7796	63.28	0.8750	2.529	0.2223	17.48	0.8600	2.885	0.3188	15.69	0.2350	1.926	0.1685	11.72	1.000
	$x_2 = 0$	16.39	0.2843	106.3	0.2350	5.306	0.1063	32.14	0.8000	1.983	0.0392	14.76	0.5200	1.798	0.0326	8.456	0.9600
	$x_2 = 10$	4.889	0.3331	29.63	0.9650	2.585	0.1341	22.33	0.8300	2.729	0.1911	16.25	0.2650	1.965	0.1176	12.41	1.000
	<i>Mean</i>	9.591	0.3852	69.31	0.6860	3.582	0.1369	23.57	0.8300	2.553	0.1385	19.18	0.3840	2.094	0.0899	11.29	0.9920
40	$x_2 = 15$	0.8645	0.0198	11.03	1.000	1.354	0.0280	7.102	1.000	1.818	0.0483	11.82	0.1400	1.106	0.0334	14.98	1.000
	$x_2 = 2$	5.091	0.2073	46.32	0.8500	0.6727	0.0216	5.507	1.000	2.552	0.0898	31.84	0.0750	0.5152	0.0170	7.361	1.000
	$x_2 = 8$	3.455	0.3626	26.91	0.8650	0.5942	0.0390	6.420	1.000	1.336	0.0973	11.16	0.1100	0.4684	0.0312	8.407	1.000
	$x_2 = 0$	2.275	0.0544	17.26	0.7800	0.4726	0.0092	4.558	1.000	1.788	0.0379	18.36	0.0750	0.3295	0.0074	6.572	1.000
	$x_2 = 10$	3.213	0.1490	15.08	0.9300	1.198	0.0709	7.836	1.000	1.908	0.0946	26.23	0.1750	0.8940	0.0465	9.424	1.000
	<i>Mean</i>	2.980	0.1586	23.32	0.8850	0.8582	0.0338	6.284	1.000	1.880	0.0736	19.88	0.1150	0.6627	0.0271	9.348	1.000

Table 3: Branin — per-level metrics, $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
10	$x_2 = 15$	25.88	0.4893	320.7	1.000	29.56	0.5406	104.7	1.000	38.97	0.7771	530.2	0.3850	38.82	0.7743	520.3	0.4250
	$x_2 = 2$	27.86	0.9856	302.1	0.8600	25.06	0.8329	209.4	0.8850	19.83	0.6723	218.5	0.4850	19.77	0.6726	182.3	0.5700
	$x_2 = 8$	15.69	1.078	169.5	0.4600	13.26	0.8785	113.3	1.000	30.26	1.812	465.1	0.0800	30.18	1.808	447.6	0.2000
	$x_2 = 0$	47.22	1.117	856.5	0.1500	41.85	0.9817	330.5	0.8450	32.82	0.7869	435.2	0.1450	32.81	0.7886	399.3	0.2000
	$x_2 = 10$	15.01	0.6307	99.37	0.9600	13.58	0.5780	113.9	0.9800	53.28	2.353	696.4	0.2400	53.22	2.351	688.1	0.3350
	<i>Mean</i>	26.33	0.8600	349.7	0.6860	24.66	0.7623	174.4	0.9420	35.03	1.280	469.1	0.2670	34.96	1.279	447.5	0.3460
25	$x_2 = 15$	6.429	0.1209	32.81	0.4750	3.802	0.0830	20.12	0.7300	4.002	0.0929	31.27	0.4050	2.343	0.0560	18.15	0.9500
	$x_2 = 2$	14.08	0.4128	110.2	0.7550	3.476	0.1285	23.53	0.8050	1.597	0.0492	9.762	0.4850	1.525	0.0405	10.01	1.000
	$x_2 = 8$	6.514	0.7329	54.17	0.9100	2.974	0.2068	16.39	0.8450	3.221	0.4004	28.72	0.5900	2.444	0.2976	13.77	1.000
	$x_2 = 0$	16.52	0.2858	107.1	0.2250	5.105	0.0989	30.53	0.7000	1.106	0.0195	4.871	0.7850	0.9591	0.0184	11.34	1.000
	$x_2 = 10$	4.426	0.2995	28.59	0.9900	2.701	0.1266	18.15	0.8700	3.920	0.2585	31.95	0.2100	2.269	0.1855	14.47	1.000
	<i>Mean</i>	9.594	0.3704	66.56	0.6710	3.611	0.1288	21.75	0.7900	2.769	0.1641	21.31	0.4950	1.908	0.1196	13.55	0.9900
40	$x_2 = 15$	0.9852	0.0209	10.74	1.000	1.192	0.0215	6.805	0.9000	0.5486	0.0107	7.218	0.4800	2.338	0.0661	15.22	1.000
	$x_2 = 2$	5.359	0.2146	47.05	0.8300	1.213	0.0495	5.608	0.9200	1.939	0.0650	26.36	0.1400	1.210	0.0435	6.854	0.9900
	$x_2 = 8$	4.016	0.3868	28.61	0.8400	0.9340	0.0695	5.523	0.9100	0.9620	0.0730	12.12	0.4150	1.251	0.0835	6.249	0.9500
	$x_2 = 0$	2.266	0.0545	18.01	0.8100	1.088	0.0316	5.163	0.9050	1.251	0.0288	15.29	0.1650	0.7429	0.0183	7.344	0.9600
	$x_2 = 10$	3.380	0.1793	18.31	0.8600	1.644	0.0956	6.347	1.000	2.209	0.1360	34.58	0.2950	1.378	0.0658	7.248	1.000
	<i>Mean</i>	3.201	0.1712	24.54	0.8680	1.214	0.0535	5.889	0.9270	1.382	0.0627	19.12	0.2990	1.384	0.0555	8.583	0.9800

Table 4: Six-hump camel — per-level metrics, $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
8	$x_2 = 0.2$	0.9745	1.360	18.57	0.0700	0.3930	0.5240	2.302	0.8250	0.4574	0.6321	6.801	0.3050	0.4581	0.6349	5.743	0.4350
	$x_2 = 0.4$	0.8588	1.087	3.892	0.8300	0.3520	0.5016	2.132	0.8650	0.6749	0.8368	6.217	0.3300	0.6301	0.7815	4.482	0.5850
	$x_2 = 0.7$	0.7935	0.8747	3.309	0.9900	0.6532	0.7363	4.491	0.6600	0.6573	0.7433	10.22	0.1700	0.6597	0.7446	9.122	0.4000
	$x_2 = 1$	1.145	1.165	11.12	0.6950	0.8244	0.7298	6.636	0.5400	0.7474	0.6216	9.631	0.0750	0.7799	0.6558	8.708	0.2550
	<i>Mean</i>	0.9429	1.122	9.221	0.6462	0.5557	0.6229	3.890	0.7225	0.6343	0.7084	8.216	0.2200	0.6319	0.7042	7.014	0.4187
20	$x_2 = 0.2$	0.1996	0.4172	1.566	0.9250	0.1277	0.2676	0.8634	0.9250	0.3616	0.5417	4.441	0.0750	0.2149	0.3769	1.226	0.8750
	$x_2 = 0.4$	0.3201	0.4179	1.517	0.9700	0.1613	0.2453	0.8614	0.8950	0.1757	0.3015	2.140	0.1900	0.1933	0.2858	1.017	0.8750
	$x_2 = 0.7$	0.1581	0.2428	1.405	0.9550	0.1421	0.2229	1.156	0.8100	0.1463	0.2360	1.541	0.3550	0.1244	0.2066	0.9197	0.9600
	$x_2 = 1$	0.2868	0.3325	1.609	0.9700	0.2250	0.2684	1.589	0.7800	0.1553	0.1843	1.205	0.4700	0.1960	0.2658	1.398	0.9650
	<i>Mean</i>	0.2411	0.3526	1.524	0.9550	0.1640	0.2511	1.118	0.8525	0.2097	0.3159	2.332	0.2725	0.1821	0.2838	1.140	0.9187
32	$x_2 = 0.2$	0.1470	0.4154	1.508	0.9100	0.0670	0.1391	0.4403	0.9700	0.1428	0.4027	1.852	0.3150	0.0944	0.2908	0.6023	1.000
	$x_2 = 0.4$	0.2014	0.3566	1.553	0.9150	0.1006	0.1612	0.6586	0.8800	0.1193	0.2678	1.530	0.2550	0.0949	0.2101	0.5612	0.9900
	$x_2 = 0.7$	0.1841	0.3277	2.077	0.7350	0.0850	0.1225	0.6137	0.8800	0.1258	0.1998	1.216	0.3800	0.1019	0.1819	0.5230	0.9850
	$x_2 = 1$	0.2206	0.2225	1.407	0.8800	0.0856	0.0924	0.5596	0.9850	0.1719	0.1944	2.101	0.4200	0.1380	0.1495	0.9745	1.000
	<i>Mean</i>	0.1883	0.3305	1.636	0.8600	0.0845	0.1288	0.5680	0.9287	0.1400	0.2662	1.675	0.3425	0.1073	0.2081	0.6652	0.9938

Table 5: Six-hump camel — per-level metrics, $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
8	$x_2 = 0.2$	0.9587	1.343	18.44	0.0500	0.4097	0.5318	2.431	0.8800	0.4820	0.6786	6.894	0.3350	0.4803	0.6727	5.995	0.4100
	$x_2 = 0.4$	0.8420	1.062	4.225	0.7350	0.3398	0.4675	1.828	0.9250	0.6352	0.7813	5.313	0.3650	0.6042	0.7421	3.980	0.5550
	$x_2 = 0.7$	0.7615	0.8419	3.532	0.9900	0.6846	0.7519	3.976	0.6750	0.6485	0.7322	9.495	0.1950	0.6589	0.7398	8.700	0.3550
	$x_2 = 1$	1.105	1.089	8.776	0.7200	0.8212	0.7122	5.270	0.5650	0.7443	0.6015	8.065	0.1300	0.7754	0.6310	6.690	0.3000
	<i>Mean</i>	0.9168	1.084	8.743	0.6238	0.5638	0.6158	3.376	0.7613	0.6275	0.6984	7.442	0.2562	0.6297	0.6964	6.341	0.4050
20	$x_2 = 0.2$	0.2209	0.4232	1.602	0.9600	0.2175	0.3302	0.9571	0.7750	0.4742	0.7717	7.023	0.1700	0.4018	0.6536	3.886	0.5750
	$x_2 = 0.4$	0.3670	0.4777	1.698	0.9400	0.1583	0.2668	1.048	0.8350	0.2371	0.4108	3.496	0.0950	0.2519	0.3973	2.006	0.6700
	$x_2 = 0.7$	0.1668	0.2468	1.363	0.9550	0.1234	0.2033	0.9071	0.9250	0.1937	0.3449	2.477	0.1350	0.1593	0.2937	1.332	0.8800
	$x_2 = 1$	0.2401	0.3872	1.896	0.9750	0.1176	0.1889	0.9879	0.9600	0.2507	0.3298	3.290	0.2000	0.2019	0.2862	1.538	0.9300
	<i>Mean</i>	0.2487	0.3837	1.640	0.9575	0.1542	0.2473	0.9751	0.8738	0.2889	0.4643	4.071	0.1500	0.2537	0.4077	2.191	0.7638
32	$x_2 = 0.2$	0.1554	0.4195	1.583	0.9000	0.0772	0.1269	0.3591	0.9300	0.1465	0.3727	1.766	0.0850	0.1004	0.1968	0.8511	1.000
	$x_2 = 0.4$	0.1826	0.3516	1.278	0.7950	0.0897	0.1462	0.5662	0.7950	0.1404	0.2041	1.828	0.1350	0.0645	0.1089	0.7446	0.9350
	$x_2 = 0.7$	0.1814	0.3164	1.794	0.7300	0.0961	0.1260	0.6441	0.7850	0.1021	0.1524	1.100	0.3400	0.0446	0.0814	0.7610	0.9850
	$x_2 = 1$	0.2744	0.3014	2.562	0.8300	0.0851	0.0856	0.5300	0.9700	0.1705	0.2213	2.294	0.2750	0.1532	0.1498	1.479	1.000
	<i>Mean</i>	0.1984	0.3472	1.804	0.8137	0.0870	0.1212	0.5249	0.8700	0.1399	0.2376	1.747	0.2088	0.0907	0.1342	0.9590	0.9800

Table 6: Rastrigin 6-D — per-level metrics, $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
80	level 1	18.25	0.7853	138.1	0.6182	17.23	0.7394	108.3	0.7021	16.31	0.7027	99.87	0.8037	16.29	0.7016	99.46	0.8027
	level 2	18.45	0.7391	122.9	0.6807	17.94	0.7258	121.6	0.7090	17.22	0.6933	102.3	0.7637	17.12	0.6890	100.4	0.7734
	level 3	21.37	0.8540	176.9	0.5439	19.71	0.8017	133.3	0.6533	20.05	0.8066	126.2	0.6846	19.94	0.8019	123.7	0.6934
	level 4	20.25	0.8813	126.5	0.7490	22.99	1.004	171.8	0.6270	21.13	0.9111	141.5	0.7051	20.56	0.8874	132.9	0.7266
	<i>Mean</i>	19.58	0.8149	141.1	0.6479	19.47	0.8178	133.7	0.6729	18.68	0.7784	117.5	0.7393	18.48	0.7700	114.1	0.7490
160	level 1	17.04	0.7346	122.5	0.6504	17.30	0.7398	95.58	0.8135	16.87	0.7162	92.62	0.8447	16.88	0.7186	93.23	0.8398
	level 2	19.91	0.8049	112.1	0.8457	21.01	0.8431	129.6	0.7441	20.94	0.8395	126.8	0.7725	21.24	0.8534	127.8	0.7686
	level 3	19.57	0.7936	110.9	0.8213	19.76	0.8060	118.7	0.7607	20.17	0.8310	119.6	0.7939	20.11	0.8285	119.3	0.7910
	level 4	19.55	0.8296	102.0	0.8418	19.26	0.8090	107.4	0.7715	20.20	0.8668	115.7	0.7871	19.99	0.8609	113.1	0.8125
	<i>Mean</i>	19.02	0.7907	111.9	0.7898	19.33	0.7995	112.8	0.7725	19.54	0.8134	113.7	0.7996	19.56	0.8154	113.4	0.8030

Table 7: Rastrigin 6-D — per-level metrics, $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
80	level 1	18.26	0.7862	138.5	0.6191	17.63	0.7541	115.2	0.6865	16.58	0.7133	101.9	0.7891	16.55	0.7121	101.5	0.7920
	level 2	18.97	0.7571	130.0	0.6719	18.41	0.7418	129.0	0.6797	17.39	0.7001	105.7	0.7520	17.35	0.6980	104.5	0.7627
	level 3	21.05	0.8430	172.8	0.5635	19.88	0.8092	140.4	0.6250	20.00	0.8043	127.6	0.6768	19.98	0.8033	126.5	0.6816
	level 4	20.65	0.8951	132.5	0.7207	23.22	1.015	181.4	0.5986	21.41	0.9253	148.2	0.6855	20.77	0.8974	138.4	0.7129
	<i>Mean</i>	19.73	0.8203	143.5	0.6438	19.78	0.8300	141.5	0.6475	18.85	0.7858	120.8	0.7258	18.66	0.7777	117.7	0.7373
160	level 1	17.11	0.7386	122.9	0.6523	17.33	0.7408	96.27	0.8066	16.95	0.7210	92.94	0.8438	16.92	0.7199	92.73	0.8438
	level 2	20.17	0.8180	114.8	0.8418	20.99	0.8436	130.3	0.7461	21.17	0.8491	128.7	0.7617	20.99	0.8419	126.3	0.7666
	level 3	19.93	0.8048	112.0	0.8096	19.78	0.8078	118.9	0.7578	20.07	0.8276	118.4	0.7979	20.02	0.8251	117.6	0.7988
	level 4	21.35	0.9165	125.6	0.7598	19.66	0.8251	110.6	0.7656	20.93	0.9002	122.3	0.7734	20.65	0.8885	118.1	0.7871
	<i>Mean</i>	19.64	0.8195	118.8	0.7659	19.44	0.8043	114.0	0.7690	19.78	0.8245	115.6	0.7942	19.65	0.8189	113.7	0.7991

Version 2 — without the Categorical GP

Table 8: Mean-over-levels metrics for every problem $\times$ budget $\times$ model (without the Categorical GP) (seed 1). Bold on green = best model in each (problem, budget, n_{rep}) block per metric (MAE/RRMSE/IS: lower; coverage: closest to 0.90).

Problem	n	Model	$n_{\text{rep}} = 10$				$n_{\text{rep}} = 3$			
			MAE $\downarrow$	RRMSE $\downarrow$	IS $\downarrow$	Cov. $\rightarrow .9$	MAE $\downarrow$	RRMSE $\downarrow$	IS $\downarrow$	Cov. $\rightarrow .9$
Branin	10	Separate GPs	26.17	0.8475	360.3	0.6680	26.33	0.8600	349.7	0.6860
		Standard LVGP	24.71	0.8273	332.6	0.3300	35.03	1.280	469.1	0.2670
		H-LVGP	43.37	1.663	589.7	0.3540	34.96	1.279	447.5	0.3460
	25	Separate GPs	9.591	0.3852	69.31	0.6860	9.594	0.3704	66.56	0.6710
		Standard LVGP	2.553	0.1385	19.18	0.3840	2.769	0.1641	21.31	0.4950
		H-LVGP	2.094	0.0899	11.29	0.9920	1.908	0.1196	13.55	0.9900
	40	Separate GPs	2.980	0.1586	23.32	0.8850	3.201	0.1712	24.54	0.8680
		Standard LVGP	1.880	0.0736	19.88	0.1150	1.382	0.0627	19.12	0.2990
		H-LVGP	0.6627	0.0271	9.348	1.000	1.384	0.0555	8.583	0.9800
Six-hump camel	8	Separate GPs	0.9429	1.122	9.221	0.6462	0.9168	1.084	8.743	0.6238
		Standard LVGP	0.6343	0.7084	8.216	0.2200	0.6275	0.6984	7.442	0.2562
		H-LVGP	0.6319	0.7042	7.014	0.4187	0.6297	0.6964	6.341	0.4050
	20	Separate GPs	0.2411	0.3526	1.524	0.9550	0.2487	0.3837	1.640	0.9575
		Standard LVGP	0.2097	0.3159	2.332	0.2725	0.2889	0.4643	4.071	0.1500
		H-LVGP	0.1821	0.2838	1.140	0.9187	0.2537	0.4077	2.191	0.7638
	32	Separate GPs	0.1883	0.3305	1.636	0.8600	0.1984	0.3472	1.804	0.8137
		Standard LVGP	0.1400	0.2662	1.675	0.3425	0.1399	0.2376	1.747	0.2088
		H-LVGP	0.1073	0.2081	0.6652	0.9938	0.0907	0.1342	0.9590	0.9800
Rastrigin 6-D	80	Separate GPs	19.58	0.8149	141.1	0.6479	19.73	0.8203	143.5	0.6438
		Standard LVGP	18.68	0.7784	117.5	0.7393	18.85	0.7858	120.8	0.7258
		H-LVGP	18.48	0.7700	114.1	0.7490	18.66	0.7777	117.7	0.7373
	160	Separate GPs	19.02	0.7907	111.9	0.7898	19.64	0.8195	118.8	0.7659
		Standard LVGP	19.54	0.8134	113.7	0.7996	19.78	0.8245	115.6	0.7942
		H-LVGP	19.56	0.8154	113.4	0.8030	19.65	0.8189	113.7	0.7991

Table 9: Branin — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
10	$x_2 = 15$	25.65	0.4851	323.4	1.000	16.74	0.3861	112.3	0.6950	52.82	0.9546	864.8	0.2250
	$x_2 = 2$	28.17	0.9843	305.5	0.8550	25.42	0.7236	238.0	0.4700	29.90	0.9110	200.5	0.6050
	$x_2 = 8$	15.29	1.050	173.9	0.4350	18.17	1.243	267.8	0.1350	53.29	3.661	855.1	0.1900
	$x_2 = 0$	47.34	1.116	897.1	0.0800	47.84	1.118	830.0	0.0000	40.37	0.9643	427.8	0.5600
	$x_2 = 10$	14.40	0.6024	101.4	0.9700	15.40	0.6668	214.7	0.3500	40.46	1.824	600.5	0.1900
	<i>Mean</i>	26.17	0.8475	360.3	0.6680	24.71	0.8273	332.6	0.3300	43.37	1.663	589.7	0.3540
25	$x_2 = 15$	5.808	0.1166	37.32	0.5700	3.470	0.0899	38.02	0.5200	3.163	0.0926	16.13	1.000
	$x_2 = 2$	14.27	0.4126	110.1	0.7850	1.699	0.0535	11.19	0.3800	1.618	0.0382	7.714	1.000
	$x_2 = 8$	6.595	0.7796	63.28	0.8750	2.885	0.3188	15.69	0.2350	1.926	0.1685	11.72	1.000
	$x_2 = 0$	16.39	0.2843	106.3	0.2350	1.983	0.0392	14.76	0.5200	1.798	0.0326	8.456	0.9600
	$x_2 = 10$	4.889	0.3331	29.63	0.9650	2.729	0.1911	16.25	0.2650	1.965	0.1176	12.41	1.000
	<i>Mean</i>	9.591	0.3852	69.31	0.6860	2.553	0.1385	19.18	0.3840	2.094	0.0899	11.29	0.9920
40	$x_2 = 15$	0.8645	0.0198	11.03	1.000	1.818	0.0483	11.82	0.1400	1.106	0.0334	14.98	1.000
	$x_2 = 2$	5.091	0.2073	46.32	0.8500	2.552	0.0898	31.84	0.0750	0.5152	0.0170	7.361	1.000
	$x_2 = 8$	3.455	0.3626	26.91	0.8650	1.336	0.0973	11.16	0.1100	0.4684	0.0312	8.407	1.000
	$x_2 = 0$	2.275	0.0544	17.26	0.7800	1.788	0.0379	18.36	0.0750	0.3295	0.0074	6.572	1.000
	$x_2 = 10$	3.213	0.1490	15.08	0.9300	1.908	0.0946	26.23	0.1750	0.8940	0.0465	9.424	1.000
	<i>Mean</i>	2.980	0.1586	23.32	0.8850	1.880	0.0736	19.88	0.1150	0.6627	0.0271	9.348	1.000

Table 10: Branin — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
10	$x_2 = 15$	25.88	0.4893	320.7	1.000	38.97	0.7771	530.2	0.3850	38.82	0.7743	520.3	0.4250
	$x_2 = 2$	27.86	0.9856	302.1	0.8600	19.83	0.6723	218.5	0.4850	19.77	0.6726	182.3	0.5700
	$x_2 = 8$	15.69	1.078	169.5	0.4600	30.26	1.812	465.1	0.0800	30.18	1.808	447.6	0.2000
	$x_2 = 0$	47.22	1.117	856.5	0.1500	32.82	0.7869	435.2	0.1450	32.81	0.7886	399.3	0.2000
	$x_2 = 10$	15.01	0.6307	99.37	0.9600	53.28	2.353	696.4	0.2400	53.22	2.351	688.1	0.3350
	<i>Mean</i>	26.33	0.8600	349.7	0.6860	35.03	1.280	469.1	0.2670	34.96	1.279	447.5	0.3460
25	$x_2 = 15$	6.429	0.1209	32.81	0.4750	4.002	0.0929	31.27	0.4050	2.343	0.0560	18.15	0.9500
	$x_2 = 2$	14.08	0.4128	110.2	0.7550	1.597	0.0492	9.762	0.4850	1.525	0.0405	10.01	1.000
	$x_2 = 8$	6.514	0.7329	54.17	0.9100	3.221	0.4004	28.72	0.5900	2.444	0.2976	13.77	1.000
	$x_2 = 0$	16.52	0.2858	107.1	0.2250	1.106	0.0195	4.871	0.7850	0.9591	0.0184	11.34	1.000
	$x_2 = 10$	4.426	0.2995	28.59	0.9900	3.920	0.2585	31.95	0.2100	2.269	0.1855	14.47	1.000
	<i>Mean</i>	9.594	0.3704	66.56	0.6710	2.769	0.1641	21.31	0.4950	1.908	0.1196	13.55	0.9900
40	$x_2 = 15$	0.9852	0.0209	10.74	1.000	0.5486	0.0107	7.218	0.4800	2.338	0.0661	15.22	1.000
	$x_2 = 2$	5.359	0.2146	47.05	0.8300	1.939	0.0650	26.36	0.1400	1.210	0.0435	6.854	0.9900
	$x_2 = 8$	4.016	0.3868	28.61	0.8400	0.9620	0.0730	12.12	0.4150	1.251	0.0835	6.249	0.9500
	$x_2 = 0$	2.266	0.0545	18.01	0.8100	1.251	0.0288	15.29	0.1650	0.7429	0.0183	7.344	0.9600
	$x_2 = 10$	3.380	0.1793	18.31	0.8600	2.209	0.1360	34.58	0.2950	1.378	0.0658	7.248	1.000
	<i>Mean</i>	3.201	0.1712	24.54	0.8680	1.382	0.0627	19.12	0.2990	1.384	0.0555	8.583	0.9800

Table 11: Six-hump camel — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
8	$x_2 = 0.2$	0.9745	1.360	18.57	0.0700	0.4574	0.6321	6.801	0.3050	0.4581	0.6349	5.743	0.4350
	$x_2 = 0.4$	0.8588	1.087	3.892	0.8300	0.6749	0.8368	6.217	0.3300	0.6301	0.7815	4.482	0.5850
	$x_2 = 0.7$	0.7935	0.8747	3.309	0.9900	0.6573	0.7433	10.22	0.1700	0.6597	0.7446	9.122	0.4000
	$x_2 = 1$	1.145	1.165	11.12	0.6950	0.7474	0.6216	9.631	0.0750	0.7799	0.6558	8.708	0.2550
	<i>Mean</i>	0.9429	1.122	9.221	0.6462	0.6343	0.7084	8.216	0.2200	0.6319	0.7042	7.014	0.4187
20	$x_2 = 0.2$	0.1996	0.4172	1.566	0.9250	0.3616	0.5417	4.441	0.0750	0.2149	0.3769	1.226	0.8750
	$x_2 = 0.4$	0.3201	0.4179	1.517	0.9700	0.1757	0.3015	2.140	0.1900	0.1933	0.2858	1.017	0.8750
	$x_2 = 0.7$	0.1581	0.2428	1.405	0.9550	0.1463	0.2360	1.541	0.3550	0.1244	0.2066	0.9197	0.9600
	$x_2 = 1$	0.2868	0.3325	1.609	0.9700	0.1553	0.1843	1.205	0.4700	0.1960	0.2658	1.398	0.9650
	<i>Mean</i>	0.2411	0.3526	1.524	0.9550	0.2097	0.3159	2.332	0.2725	0.1821	0.2838	1.140	0.9187
32	$x_2 = 0.2$	0.1470	0.4154	1.508	0.9100	0.1428	0.4027	1.852	0.3150	0.0944	0.2908	0.6023	1.000
	$x_2 = 0.4$	0.2014	0.3566	1.553	0.9150	0.1193	0.2678	1.530	0.2550	0.0949	0.2101	0.5612	0.9900
	$x_2 = 0.7$	0.1841	0.3277	2.077	0.7350	0.1258	0.1998	1.216	0.3800	0.1019	0.1819	0.5230	0.9850
	$x_2 = 1$	0.2206	0.2225	1.407	0.8800	0.1719	0.1944	2.101	0.4200	0.1380	0.1495	0.9745	1.000
	<i>Mean</i>	0.1883	0.3305	1.636	0.8600	0.1400	0.2662	1.675	0.3425	0.1073	0.2081	0.6652	0.9938

Table 12: Six-hump camel — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
8	$x_2 = 0.2$	0.9587	1.343	18.44	0.0500	0.4820	0.6786	6.894	0.3350	0.4803	0.6727	5.995	0.4100
	$x_2 = 0.4$	0.8420	1.062	4.225	0.7350	0.6352	0.7813	5.313	0.3650	0.6042	0.7421	3.980	0.5550
	$x_2 = 0.7$	0.7615	0.8419	3.532	0.9900	0.6485	0.7322	9.495	0.1950	0.6589	0.7398	8.700	0.3550
	$x_2 = 1$	1.105	1.089	8.776	0.7200	0.7443	0.6015	8.065	0.1300	0.7754	0.6310	6.690	0.3000
	<i>Mean</i>	0.9168	1.084	8.743	0.6238	0.6275	0.6984	7.442	0.2562	0.6297	0.6964	6.341	0.4050
20	$x_2 = 0.2$	0.2209	0.4232	1.602	0.9600	0.4742	0.7717	7.023	0.1700	0.4018	0.6536	3.886	0.5750
	$x_2 = 0.4$	0.3670	0.4777	1.698	0.9400	0.2371	0.4108	3.496	0.0950	0.2519	0.3973	2.006	0.6700
	$x_2 = 0.7$	0.1668	0.2468	1.363	0.9550	0.1937	0.3449	2.477	0.1350	0.1593	0.2937	1.332	0.8800
	$x_2 = 1$	0.2401	0.3872	1.896	0.9750	0.2507	0.3298	3.290	0.2000	0.2019	0.2862	1.538	0.9300
	<i>Mean</i>	0.2487	0.3837	1.640	0.9575	0.2889	0.4643	4.071	0.1500	0.2537	0.4077	2.191	0.7638
32	$x_2 = 0.2$	0.1554	0.4195	1.583	0.9000	0.1465	0.3727	1.766	0.0850	0.1004	0.1968	0.8511	1.000
	$x_2 = 0.4$	0.1826	0.3516	1.278	0.7950	0.1404	0.2041	1.828	0.1350	0.0645	0.1089	0.7446	0.9350
	$x_2 = 0.7$	0.1814	0.3164	1.794	0.7300	0.1021	0.1524	1.100	0.3400	0.0446	0.0814	0.7610	0.9850
	$x_2 = 1$	0.2744	0.3014	2.562	0.8300	0.1705	0.2213	2.294	0.2750	0.1532	0.1498	1.479	1.000
	<i>Mean</i>	0.1984	0.3472	1.804	0.8137	0.1399	0.2376	1.747	0.2088	0.0907	0.1342	0.9590	0.9800

Table 13: Rastrigin 6-D — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
80	level 1	18.25	0.7853	138.1	0.6182	16.31	0.7027	99.87	0.8037	16.29	0.7016	99.46	0.8027
	level 2	18.45	0.7391	122.9	0.6807	17.22	0.6933	102.3	0.7637	17.12	0.6890	100.4	0.7734
	level 3	21.37	0.8540	176.9	0.5439	20.05	0.8066	126.2	0.6846	19.94	0.8019	123.7	0.6934
	level 4	20.25	0.8813	126.5	0.7490	21.13	0.9111	141.5	0.7051	20.56	0.8874	132.9	0.7266
	<i>Mean</i>	19.58	0.8149	141.1	0.6479	18.68	0.7784	117.5	0.7393	18.48	0.7700	114.1	0.7490
160	level 1	17.04	0.7346	122.5	0.6504	16.87	0.7162	92.62	0.8447	16.88	0.7186	93.23	0.8398
	level 2	19.91	0.8049	112.1	0.8457	20.94	0.8395	126.8	0.7725	21.24	0.8534	127.8	0.7686
	level 3	19.57	0.7936	110.9	0.8213	20.17	0.8310	119.6	0.7939	20.11	0.8285	119.3	0.7910
	level 4	19.55	0.8296	102.0	0.8418	20.20	0.8668	115.7	0.7871	19.99	0.8609	113.1	0.8125
	<i>Mean</i>	19.02	0.7907	111.9	0.7898	19.54	0.8134	113.7	0.7996	19.56	0.8154	113.4	0.8030

Table 14: Rastrigin 6-D — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
80	level 1	18.26	0.7862	138.5	0.6191	16.58	0.7133	101.9	0.7891	16.55	0.7121	101.5	0.7920
	level 2	18.97	0.7571	130.0	0.6719	17.39	0.7001	105.7	0.7520	17.35	0.6980	104.5	0.7627
	level 3	21.05	0.8430	172.8	0.5635	20.00	0.8043	127.6	0.6768	19.98	0.8033	126.5	0.6816
	level 4	20.65	0.8951	132.5	0.7207	21.41	0.9253	148.2	0.6855	20.77	0.8974	138.4	0.7129
	<i>Mean</i>	19.73	0.8203	143.5	0.6438	18.85	0.7858	120.8	0.7258	18.66	0.7777	117.7	0.7373
160	level 1	17.11	0.7386	122.9	0.6523	16.95	0.7210	92.94	0.8438	16.92	0.7199	92.73	0.8438
	level 2	20.17	0.8180	114.8	0.8418	21.17	0.8491	128.7	0.7617	20.99	0.8419	126.3	0.7666
	level 3	19.93	0.8048	112.0	0.8096	20.07	0.8276	118.4	0.7979	20.02	0.8251	117.6	0.7988
	level 4	21.35	0.9165	125.6	0.7598	20.93	0.9002	122.3	0.7734	20.65	0.8885	118.1	0.7871
	<i>Mean</i>	19.64	0.8195	118.8	0.7659	19.78	0.8245	115.6	0.7942	19.65	0.8189	113.7	0.7991